

No.	Solution
6(a)	From the graph, R at $20^{\circ}\text{C} = 60\ \Omega$ By potential divider principle, $0.43 = \frac{r}{r + 60} \times 1.50$ $0.43(r + 60) = 1.50r$ $1.07r = 25.8$ $r = \frac{25.8}{1.07} = 24.112 = 24\ \Omega \text{ (shown)}$
6(b)	From the graph, R at $32^{\circ}\text{C} = 40\ \Omega$ $R_{\text{eff}} = \left(\frac{1}{24} + \frac{1}{40} \right)^{-1} = 15\ \Omega$ $I = \frac{\varepsilon}{R_{\text{eff}}} = \frac{1.50}{15} = 0.10\ \text{A}$